

# Great Gains in Mass Spectrometry Data Interpretation

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For about three decades, the analysis of mass spectrometry (MS) data has allowed large-scale amino acid sequence identification and protein quantification.<sup>1</sup> Over these decades, numerous algorithmic advances have steadily improved MS data interpretation for sequence identification.<sup>2</sup> The pace of this progress continues unabated along many avenues, including improved interpretation of complex spectra from multiplexed data independent acquisition (plexDIA),<sup>3</sup> *de novo* sequencing with new embeddings and transformer neural network architectures,<sup>4</sup> improvements in open searches and the identification of peptide modifications,<sup>5</sup> and improved models of isotopic compositions.

The latter advances are exemplified by a new approach termed Conditional fragment Ion Distribution Search (CIDS).<sup>6</sup> CIDS can substantially increase sequence identification rates for peptides labeled by using heavy water (<sup>2</sup>H) or <sup>15</sup>N since such peptides have structural isomers and the distributions of their fragment ions have been difficult to predict. CIDS predicts the isotopologue distributions of MS2 fragment ions, which are then searched by Comet.<sup>7</sup> This CIDS-Comet approach improved peptide identification by over 100% compared to conventional database search methods that do not model structural isomers and their corresponding fragments.<sup>6</sup> CIDS achieves these large gains while using simplifying approximations, thus highlighting the opportunities for major gains in the field.

This individual example highlights a general point: MS collects large data sets of highly accurate and precise measurements, and improvements in their interpretation offer opportunities for great gains. These opportunities are particularly prominent in challenging contexts, such as single-cell proteomics,<sup>8</sup> intact protein analysis, and plexDIA;<sup>9</sup> indeed, almost any MS data analysis workflow can be substantially improved, sometimes by over 100% as demonstrated by CIDS.<sup>6</sup>

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